



Universidad del Desarrollo

Overtime Among Fixed-Term Staff of the Servicio de Salud Metropolitano Central

Exploring Two Candidate Lines of Inquiry
WI · Discovering Questions Worth Investigating

Group 01

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Abstract

This report explores two candidate lines of inquiry on the use of overtime among fixed-term (*contrata*) staff of the Servicio de Salud Metropolitano Central (SSMC), in order to select one for the C1 project. The data source is the set of monthly payrolls published under Active Transparency between September 2024 and August 2026 (44,777 records). Candidate A seeks to identify overtime payments that exceed the legal limit of 40 daytime hours per month or that are atypical for the position and staff category. Candidate B seeks to identify positions and staff categories with persistent overtime use, as a possible signal of a staffing gap. The data were processed in Python with explicit quality checks, which showed that each row is a contract record rather than a person, that there is no unique identifier, and that January 2025 and December 2025 contain atypical records. In Candidate A, 182 person-months exceed the limit, 99 of them in January 2025. In Candidate B, 20.5% of staff present for at least 12 months show persistent use, concentrated among auxiliary and administrative staff. The team provisionally prefers Candidate B; its main uncertainty is that the source does not report authorized staffing, so it cannot demonstrate a gap.

Keywords: overtime; Active Transparency; health services; human resources management; Servicio de Salud Metropolitano Central.

Contents

Abstract	1
Introduction	3
1. Candidate Lines of Inquiry	4
1.1 Organization, Data Source and Scope	4
1.2 Candidate A: Overtime Above the Legal Limit and Atypical Cases	5
1.2.1 Problem	5
1.2.2 Relevance	5
1.2.3 Tentative Question and Proposed Use	6
1.2.4 Data Opportunity	7
1.3 Candidate B: Persistent Overtime as a Signal of a Staffing Gap	7
1.3.1 Problem	7
1.3.2 Relevance	7
1.3.3 Tentative Question and Proposed Use	8
1.3.4 Data Opportunity	8
1.4 Differences Between the Candidates	9
2. Data Inspection	9
2.1 Source and Procedure	10
2.2 Unit of Observation, Fields, Period and Coverage	11
2.3 Initial Quality Checks	12
2.4 First Observations for Candidate A	14
2.5 First Observations for Candidate B	16
2.6 Proposed and Verified Joins	18
3. Critical Comparison of the Candidates	18
3.1 Comparison Matrix	18
3.2 Provisional Preference	19
3.3 Questions for Feedback	19
4. Future Machine Learning and Dashboard Potential	20
4.1 Candidate A	20
4.2 Candidate B	20
5. Conclusions	20
6. Process Record	21
6.1 Decisions and Checks	21
6.2 Instructor Feedback and Response	22

6.3 Individual Contributions	22
6.4 Use of Artificial Intelligence and Verification.....	22
6.5 Individual Reflection.....	23
References.....	23
Appendices.....	25
Appendix A. Field Dictionary of the Fixed-Term Staff Payroll	26
Appendix B. Supplementary Inspection Tables	26

Introduction

Overtime is a legitimate mechanism in public administration for covering tasks that cannot be postponed. In health services, where many functions require continuous care, overtime is used frequently and its control is especially relevant: a payment without support affects the use of public resources, and a persistent reliance on additional hours may indicate that staffing is insufficient. A recent audit by the Contraloría General de la República (Office of the Comptroller General, CGR) of a public hospital, reviewed in Section 1, shows that both aspects are a matter of concern in Chile.

The objective of this report is to present and compare two candidate lines of inquiry on the use of overtime among fixed-term staff of the SSMC, based on the data published under Active Transparency. For each line, we define the problem, review the documentary evidence that supports it, formulate a tentative question and its possible use, and examine what the available data can answer. We then inspect the actual data to assess the feasibility of each line, compare both alternatives and state a provisional preference, together with the questions on which we request feedback before developing the C1 project.

The report is organized as follows. Section 1 presents the two candidate lines and their documentary foundation. Section 2 describes the data inspection and its first observations. Section 3 compares the candidates and sets out the provisional preference and the questions for feedback. Section 4 discusses the potential of each line for a future machine learning and dashboard extension. Section 5 presents the conclusions, Section 6 contains the Process record, and the references and appendices follow.

1. Candidate Lines of Inquiry

1.1 Organization, Data Source and Scope

The topic of this work is the use of overtime by the staff of a public health service. The organization studied is the Servicio de Salud Metropolitano Central (SSMC). It is part of the Ministry of Health (MINSAL), reports to the Undersecretariat of Healthcare Networks and is one of the country's 29 health services. It has a staff of 4,600 employees and serves an assigned population of more than one million people in the municipalities of Santiago, Estación Central, Cerrillos, Maipú and Pedro Aguirre Cerda (SSMC, n.d.-a). Its primary care network includes municipally administered centers and centers administered directly by the Service (SSMC, n.d.-b), and it also includes SAMU Metropolitano, which handles calls to the 131 emergency number throughout the Metropolitan Region (SSMC, n.d.-c).

The data source is the payroll of SSMC fixed-term staff published under Active Transparency (*Transparencia Activa*) through the Transparency Portal, organization code AO006 (SSMC, 2024–2026). For each record, this payroll reports the staff category (*estamento*), position or function, grade or working hours, gross and net pay for the month, entitlement to overtime, and the amount and number of daytime, night and holiday overtime hours, together with the start and

end dates of the contract. Because it covers only fixed-term staff, the source does not include permanent (*planta*) staff, fee-based (*honorarios*) staff or staff hired under the Labor Code. It therefore does not represent the Service's entire workforce, and any result must be understood as referring to this segment of staff.

Regarding the employment regime, health service staff are generally governed by the Administrative Statute (Law 18,834). Professionals under Laws 15,076 and 19,664 have special statutes, to which the Administrative Statute applies supplementarily (FALMED, n.d.). Law 18,834 allows overtime work to be ordered after the ordinary working day, at night, or on Saturdays, Sundays and holidays, when tasks that cannot be postponed must be completed. This work is compensated with complementary rest or, if that is not possible for reasons of good service, with a pay surcharge (Law 18,834, arts. 66 and 69, cited in Contraloría General de la República [CGR], 2026).

It is important to state the scope of both candidates. The available data correspond to overtime payments and show how many hours were paid, **but not whether they were actually worked**. For this reason, neither candidate seeks to establish the existence of fraud; instead, both seek to identify atypical or persistent patterns that justify further review. In addition, although the information is public, this work does not identify individual employees: results are presented in aggregate form by position, staff category and number of cases.

1.2 Candidate A: Overtime Above the Legal Limit and Atypical Cases

1.2.1 Problem

The first candidate addresses the concentration of overtime payments that exceed the monthly legal limit, or that are atypical relative to the employee's position and staff category, among SSMC fixed-term staff between September 2024 and August 2026.

1.2.2 Relevance

Overtime pay in public administration is subject to an explicit limit. Article 9 of Law 19,104 establishes that the maximum number of daytime overtime hours whose payment may be authorized is 40 hours per employee per month, and that any excess that cannot be paid in money must be compensated with rest (Law 19,104, 1991; FALMED, n.d.). This limit refers only to the payment of daytime hours, so it does not cover night hours, hours worked on Saturdays, Sundays or holidays, or hours the service compensates with rest (FALMED, n.d.). In turn, the Ministry of Finance may, by decree, exempt services that, due to special circumstances, require certain staff to work more hours (CGR, Opinion No. 45,806 of 2009, cited in vLex, n.d.). For professionals under Law 15,076 and staff under Law 19,664, Law 18,834 sets no maximum number of overtime hours (FALMED, n.d.), so the medical staff category must be analyzed separately.

Health services are also subject to a specific restriction. Article 95 of Decree with Force of Law (DFL) No. 1 of 2005 of the Ministry of Health provides that staff who receive a shift allowance (*asignación de turno*) may not perform overtime work of any kind, except for

unforeseen work caused by health emergencies or urgent patient care needs, as determined by the director of the facility through a reasoned resolution (CGR, 2026).

The relevance of this problem has become evident in recent audits of the sector. The Regional Comptroller's Office of La Araucanía, in its Final Special Investigation Report No. 70 of 2026, examined overtime at Hospital San José de Victoria between January 2024 and June 2025, following a complaint. In that period, 1,000 employees received overtime payments totaling CLP 2,086,479,906, of which CLP 1,416,186,807 corresponded to 2024 and CLP 670,293,099 to the first half of 2025 (CGR, 2026).

The investigation found that the administrative acts ordering the overtime were regularized between 54 and 610 calendar days after the work was performed, without identifying the non-postponable tasks that justified it. It also found that third- and fourth-shift staff performed overtime worth CLP 562,964,119 despite the restriction in Article 95 of DFL No. 1, and that payments of CLP 3,784,295 were made without support in attendance records or with calculation errors. The Comptroller warned that errors and omissions in attendance records undermine confidence in the data and may lead to payments for days on which the employee did not work (CGR, 2026).

This report also provides a direct methodological precedent for this candidate. To select the cases to review, the Comptroller used payroll data and included in its sample, alongside the employees named in the complaint, those who recorded more than 2,500 overtime hours and more than CLP 10,000,000 paid during the period, and then compared those cases with attendance records (CGR, 2026). The present candidate replicates the first stage of that logic: it uses payment data to prioritize cases, acknowledging that verifying them requires attendance information that is not public.

There are also precedents in the municipal sector. The Consejo para la Transparencia (Council for Transparency, 2022) characterized municipal overtime using data published on the Transparency Portal, and the Comptroller compiled a consolidated dataset of municipal overtime for 2024 in order to identify behavior outside national averages (Inteligencia Jurídica, 2025). Although these studies do not refer to health services, they support the use of transparency data for this type of analysis.

Regarding its relationship with Ministry of Health policies, this candidate is not linked to a health priority, and this work does not force that match. Its relevance lies in public management, specifically expenditure control and regulatory compliance within a health service. Note that the 40-hour daytime limit is a legal rule, not a policy target; Active Transparency data do make it possible to measure paid daytime hours against that limit.

1.2.3 Tentative Question and Proposed Use

The tentative question, formulated as this team's proposal, is: what proportion of SSMC fixed-term staff record payments for more than 40 daytime overtime hours in a month, and in which staff categories and positions are cases of atypically high night overtime concentrated?

As this team's proposal, the user of this analysis would be the SSMC internal audit unit, which could use it to decide which overtime authorizations to review first in the Service's Directorate, in its directly administered primary care facilities and in SAMU Metropolitano.

1.2.4 Data Opportunity

The monthly payroll of SSMC fixed-term staff makes it possible to answer the descriptive part of the question, because it reports the daytime, night and holiday overtime hours paid to each record, together with its staff category, position and grade.

However, the source has important gaps. It is not possible to verify that the hours paid were worked, because attendance records are not public and, as the Victoria case shows, the Comptroller itself had to rely on them to do so (CGR, 2026). The special allowances column records numeric codes, such as 186, 204 or 202, whose meaning is not documented in the file; it remains to be verified whether any of them corresponds to the shift allowance, which is needed to apply the restriction in Article 95 of DFL No. 1. Nor is it known whether any Ministry of Finance decree exempts the SSMC from the limit. The payment month may not coincide with the month in which the hours were worked; in the Victoria case, hours from one month were paid in the following month's payroll (CGR, 2026). The payroll does not report the facility or unit where each employee works, and the position or function column does not allow it to be identified (Section 2.2). Finally, permanent staff are not included.

The proposed period also involves two regulatory considerations. In health services, overtime does not constitute permanent pay and is not received during vacations, sick leave or other leave (DFL No. 1 of 2005, art. 97, cited in CGR, 2026). In addition, the Comptroller clarified that overtime may be paid only for work actually performed, a criterion applicable from July 19, 2025 (CGR, Opinion No. E121666 of 2025, cited in Diario Constitucional, 2025). Because the study period spans that date, comparisons between months must take this change into account.

1.3 Candidate B: Persistent Overtime as a Signal of a Staffing Gap

1.3.1 Problem

The second candidate addresses the persistent, month-to-month use of overtime and replacement contracts to cover clinical and support functions among SSMC fixed-term staff between September 2024 and August 2026. Particular attention is paid to emergency care staff, given that SAMU Metropolitano provides pre-hospital care to the entire region and transports critical patients (SSMC, 2025). Structural use of overtime in these functions could indicate a staffing gap that is being covered with additional hours rather than with new staff.

1.3.2 Relevance

The Hospital San José de Victoria case shows that overtime use in a health facility can be widespread rather than exceptional: 1,000 employees received it over an 18-month period, and 332 of them received a shift allowance (CGR, 2026). After the report was published, the management of the Servicio de Salud Araucanía Norte explained that these hours responded to the need to

ensure continuity of care (BioBioChile, 2026). The hospital's management added that the aim was to cover unforeseen absences in a shift system that requires 24-hour coverage (Las Noticias de Malleco, 2026). These statements suggest that overtime may operate as a mechanism to make up for staffing gaps.

The international literature warns that this mechanism has costs. A systematic review concludes that error rates appear to be higher among nurses working shifts of 12 hours or more in acute care hospitals (Clendon & Gibbons, 2015). A more recent rapid review links understaffing to pressure to work longer hours and finds that shifts longer than 12 hours increase the risk of occupational fatigue (Scott-Marshall, 2024). The same review cites work by Olds and Clarke according to which overtime increased the risk of needlestick injuries by 20% and of medication errors by 30% (Scott-Marshall, 2024). This evidence comes from other countries and focuses on nursing staff, so it is used to support the concern, not to estimate its effects at the SSMC.

Regarding Ministry of Health policies, the National Health Strategy for the 2030 Health Objectives (MINSAL, 2022) identifies, among the problems of the network, high territorial inequity in the distribution of human resources, deficiencies in selection and recruitment processes, and insufficient policies on staff safety and good treatment. A subsequent review of the Strategy, carried out between 2022 and 2024, redefined its structure, priority topics, impact objectives and indicators (MINSAL, 2026). The Strategy does not set a target for overtime. Nor do Active Transparency data make it possible to measure the problems it diagnoses, such as territorial inequity or staff safety; they capture only one possible symptom of them: the recurring use of overtime. Therefore, this work does not measure compliance with an official target.

1.3.3 Tentative Question and Proposed Use

The tentative question, formulated as this team's proposal, is: which positions and staff categories of SSMC fixed-term staff show persistent overtime payments throughout the period, and what is the weight of that expenditure relative to their gross pay?

As this team's proposal, the users of this analysis would be the SSMC people management and development units, present in the Service's Directorate and in the Primary Care Directorate (SSMC, n.d.-b). They could use it to decide in which positions it would be advisable to increase staffing, rather than sustaining recurring overtime payments.

1.3.4 Data Opportunity

The monthly payroll series makes it possible to follow records over time and observe the persistence of overtime by position and staff category. The observations column mentions replacement or substitution in 1,921 records (4.3%); this figure comes from a text search, not a case-by-case review. The start and end dates make it possible to distinguish short-term appointments. This candidate is particularly relevant for fixed-term staff, since replacements are recorded under this type of appointment.

The source nevertheless has important gaps. It does not report authorized staffing or vacancies, which would be needed to measure the gap directly, nor the facility or unit of each employee, which prevents separating SAMU, primary care or Directorate staff directly. It does not record sick leave or vacations either; because overtime is not paid during those periods (DFL No. 1 of 2005, art. 97, cited in CGR, 2026), a month without overtime could reflect an absence rather than a lower need. The payroll publishes names but not national ID numbers (RUT), and the same person may hold more than one contract, which makes individual tracking difficult. Because permanent staff are not included, it is not possible to know whether the overtime burden is distributed differently between permanent and fixed-term staff. Finally, a possible join with other course datasets, such as Urgencias or the Monthly Statistical Summaries (REM), remains a proposal and has not been verified.

1.4 Differences Between the Candidates

Both candidates use the same data source, which the brief allows, but they answer different questions and support different decisions. Candidate A poses a control question: it seeks to identify which payments should be reviewed, focuses on each month separately and on regulatory thresholds, and its user would be internal audit. Its logic follows the first stage of the method applied by the Comptroller in the Hospital San José de Victoria case. Candidate B, by contrast, poses a planning question: it seeks to identify where staff may be lacking, requires the full series for the period because it focuses on persistence, and its user would be the people management area. Table 1 frames both candidates using the seven steps from Class 03.

Table 1

Framing of the Candidates Using the Seven Steps from Class 03

Step	Candidate A	Candidate B
Decision	Which overtime authorizations to review first	In which staff categories to assess increasing staffing
Success criterion (proposed)	Fewer payments above the limit without an authorized exception	Lower recurring reliance on overtime
Business question	Section 1.2.3	Section 1.3.3
Business process	Authorization → work performed → attendance record → payroll payment → publication	Same process, observed across several months
Entities, events and states	Employee (no unique ID), contract, monthly payment	Employee, contract, monthly sequence of payments
Data representation	Row = contract record per month; person identified by name	Same as Candidate A
Measurement	Paid daytime hours per person-month above 40	Share of months with overtime $\geq 75\%$; overtime share of gross pay

Note. Success criteria and users are proposals of this team.

2. Data Inspection

2.1 Source and Procedure

The data come from the State Transparency Portal, Active Transparency section of the SSMC (organization code AO006), specifically from the fixed-term staff payroll. The 24 monthly files between September 2024 and August 2026 were downloaded in CSV format on September 15, 2026. Each file corresponds to one month of pay.

The analysis was carried out in Python, in a Jupyter notebook (analysis/EDA.ipynb), using the pandas, NumPy, Matplotlib and Seaborn libraries.

The files were read with a semicolon separator and latin-1 encoding, since they are not UTF-8 encoded. All columns were read as text, to prevent the program from assigning data types automatically. This decision responds to the fact that several columns combine formats: for example, grade or working hours includes numeric values and values such as "44 Hrs.", and amounts include the peso sign. In this way, every subsequent conversion is made explicitly.

Each file contains 21 columns with information and a final unnamed column, generated because each line ends with a semicolon. Before dropping this column, we verified that it was empty in all 24 files. A variable with the year and month of the source file was added to each record, and the 24 files were combined into a single table. We checked that the number of rows in the combined table equaled the sum of the rows of the individual files: 44,777 records (notebook, Section 1). Table 2 shows the number of records per file; the number of monthly records ranged from 1,781 in September 2024 to 1,951 in February 2026.

Table 2

Records per Monthly File, SSMC Fixed-Term Staff Payroll, September 2024 – August 2026

Year	Month	Records	Year	Month	Records
2024	September	1,781	2025	September	1,878
2024	October	1,799	2025	October	1,882
2024	November	1,798	2025	November	1,919
2024	December	1,803	2025	December	1,918
2025	January	1,825	2026	January	1,907
2025	February	1,842	2026	February	1,951
2025	March	1,845	2026	March	1,908
2025	April	1,861	2026	April	1,903
2025	May	1,874	2026	May	1,886
2025	June	1,870	2026	June	1,865
2025	July	1,872	2026	July	1,857
2025	August	1,884	2026	August	1,849

Note. Total: 44,777 records. Own elaboration based on the SSMC fixed-term staff payroll, Active Transparency (notebook, Section 1).

Next, the variables needed for the analysis were prepared. Pay amounts, recorded as text with the peso sign and a period as thousands separator, were converted to integers in Chilean pesos. Gross and net pay were converted without errors. In contrast, the additional pay and bonus columns contained values recorded with a dash ("-") instead of an amount, concentrated in specific months. Because the source does not indicate whether the dash means no amount or unreported information, these values were treated as missing rather than zero (notebook, Section 2.1).

The daytime, night and holiday overtime columns combine the amount paid and the number of hours in a single text (for example, "\$ 154.547 : 40,00 hrs"), or state "No tiene" (none) when there were no hours of that type in the month. Each column was split into two numeric variables: the amount, in pesos, and the number of hours. Records with "No tiene" were coded as zero, since they indicate the absence of paid overtime rather than missing data (notebook, Section 2.2).

Finally, contract start and end dates were converted to date format. Although the header of these columns indicates the format "dd/mm/aa", the values record the year with four digits, so the conversion used the full day/month/year format. All dates were converted without errors (notebook, Section 2.3).

2.2 Unit of Observation, Fields, Period and Coverage

Each file corresponds to one month's payroll and each row to a record in that payroll. The period covers 24 consecutive months, from September 2024 to August 2026. Coverage is limited to SSMC fixed-term staff; it does not include permanent, fee-based or Labor Code staff. The payroll does not contain a unique identifier per employee, such as the RUT; the only data that allow people to be distinguished is the full name.

Table 3 presents the payroll fields relevant to the candidates and the use each candidate makes of them. The complete field dictionary is provided in Appendix A.

Table 3
Payroll Fields Relevant to the Candidates

Field	Content	Cand. A	Cand. B
Year and month	Pay period	Yes	Yes
Staff category (<i>estamento</i>)	Professional, technical, administrative, auxiliary, or physicians and related	Yes	Yes
Position or function	Declared function of the record	Yes	Yes
Grade (EUS) or working hours	Grade on the pay scale or contracted weekly hours	Yes	Yes
Gross pay for the month	Includes bonuses, special allowances and overtime	No	Yes
Entitlement to overtime	Whether the record is entitled to overtime	Yes	Yes
Daytime overtime	Amount paid and number of daytime hours	Yes	Yes
Night overtime	Amount paid and number of night hours	Yes	Yes
Holiday overtime	Amount paid and number of holiday hours	Yes	Yes

Field	Content	Cand. A	Cand. B
Start and end dates	Validity of the contract	No	Yes
Observations	Free text about the month's payment	Yes	Yes

Note. Own elaboration based on the SSMC fixed-term staff payroll, Active Transparency.

The position or function column records 95 distinct values, which drop to 81 when upper and lower case are unified. The 15 most frequent values, covering 95.7% of records, correspond to generic functions, such as clinical care, administrative support or driver, and include synonyms such as *conductor* and *chofer* (both meaning driver). Therefore, this column does not identify the facility, and its use for analysis by position requires normalizing it, which is left for C1 (notebook, Section 2.4).

To determine the unit of observation, we counted how many times each name appears within the same month. In all 24 months there were people with more than one record in the same month: between 57 and 68 people per month, generating between 62 and 76 additional rows, equivalent to between 3.4% and 4.0% of monthly rows. One person had up to six rows in a single month (Appendix B, Table B1). Therefore, each row of the payroll corresponds to a contract or appointment record in force in the month, and not necessarily to a distinct person (notebook, Section 3).

In total, 2,430 distinct people appear in the payroll over the 24 months, more than in any individual month, which ranged from 1,717 to 1,880 people. This difference indicates turnover among fixed-term staff during the period.

This feature has consequences for both candidates. In Candidate A, the legal limit of 40 daytime overtime hours applies per employee per month (Law 19,104, 1991), so hours must be summed per person before comparing them with the limit. In Candidate B, the persistence of overtime must be measured per person across months. Because there is no unique identifier such as the RUT, people are identified by their full name. This prevents distinguishing different people with the same name and may split one person into two if their name is recorded differently across months.

2.3 Initial Quality Checks

During data preparation, quality checks were applied with two purposes. The first was to verify that loading and conversions did not lose or alter information. The second was to identify features of the source that condition the analysis of the candidates. Table 4 summarizes all checks and classifies them according to the Five Cs of data quality: clean, consistent, complete, current and collaborative.

The loading checks confirmed that the dropped final column was empty in all 24 files and that combining the files preserved all 44,777 records. The conversions of gross and net pay, of daytime and night overtime and of start and end dates were completed without errors; in these columns, all values had the expected format. In contrast, the additional pay and bonus columns

contained values recorded with a dash instead of an amount. In the first, the dashes cover all records for January 2025. In the second, they are concentrated in May 2025, July 2026, December 2024 and November 2025. Because the source does not clarify whether the dash indicates no amount or unreported information, these values were treated as missing. Neither of these two columns is used directly in the candidates.

The checks also revealed five features of the source that condition the analysis. First, the holiday overtime column records no values in any of the 24 months, so the analysis of hours is limited to daytime and night hours. Second, each row of the payroll corresponds to a contract or appointment record and not to a person: every month there were people with more than one record, between 57 and 68 per month, with a maximum of six records for one person. Third, the payroll does not include a unique identifier such as the RUT, so identifying people depends on the full name.

Fourth, the staff category column behaves anomalously in December 2025. Comparing each person's staff category between consecutive months, 218 people change category in December 2025 relative to the previous month, and 213 change again in January 2026. In the remaining months, the number of changes does not exceed six (Appendix B, Table B2). This pattern, in which the same people change category for a single month and then return to the original, suggests a recording error in the December 2025 file. Because the correct category cannot be determined, that month was excluded from analyses that depend on staff category and kept in the others.

Finally, January 2025 behaves differently from the other months. In addition to recording dashes in the entire additional pay column, it concentrates 99 of the 182 cases exceeding the daytime overtime limit in the period (Section 2.4). The source does not make it possible to determine whether this concentration reflects a real operational situation or a different way of recording payments in that file.

Three additional checks were applied to fields not used in the initial summaries (notebook, Section 2.4): the values of the position column (Section 2.2), declared overtime entitlement against paid hours (Section 2.4), and the special allowance codes, which have no legend in the file (Section 1.2.4).

Table 4
Quality Checks Applied to the SSMC Fixed-Term Staff Payroll

Check	Purpose	Result	Five Cs
Content of the final unnamed column	Verify that dropping it loses no information	Empty in all 24 files	Clean
Rows before and after combining files	Verify that combining neither loses nor duplicates records	44,777 in both cases	Complete
Conversion of gross and net pay	Detect unexpected formats	No non-convertible values	Clean

Check	Purpose	Result	Five Cs
Conversion of additional pay	Detect unexpected formats	1,825 "-" values, all records for January 2025	Complete
Conversion of bonuses	Detect unexpected formats	3,065 "-" values: May 2025 (1,507), July 2026 (1,479), December 2024 (70), November 2025 (9)	Complete
Format of overtime columns	Verify that all values can be split into amount and hours	All in the expected format or "No tiene"; no blanks or other formats	Clean
Holiday overtime	Verify availability of the information	No holiday hours in any of the 44,777 records	Complete
Conversion of start and end dates	Detect blank or unexpected dates	No errors; header says "dd/mm/aa" but values use a four-digit year	Collaborative
Repeated names within a month	Determine the unit of observation	57–68 people per month with more than one record (maximum six); the unit is the contract record	Consistent
Person identifier	Assess whether people can be tracked across months	No RUT; the name is used, with risk of homonyms and spelling variations	Consistent
Staff category changes between consecutive months	Verify consistency of the staff category column	December 2025: 218 changes; January 2026: 213; other months: 0–6. December 2025 excluded from analyses by category	Consistent
Monthly distribution of cases above 40 daytime hours	Detect atypical months	January 2025 concentrates 99 of 182 cases (54%)	Consistent
Overtime amount versus gross pay	Verify consistency between amounts	9 person-months with overtime above gross pay (maximum 451.6%); 3 with gross pay equal to zero	Clean
Values of position or function	Verify consistency of labels	95 values; 81 after unifying case; synonyms (conductor/chofer)	Consistent
Declared entitlement versus paid hours	Verify consistency between fields	207 records without entitlement but with paid hours; 407 with entitlement and no hours	Consistent
Mentions in observations	Verify availability of replacement information	1,921 records mention replacement or substitution	Complete
Special allowance codes	Verify that codes can be interpreted	Numeric codes with no legend in the file	Collaborative
Timeliness of the data	Verify that the period fits the decision window	Data through August 2026, downloaded September 15, 2026; payment month may differ from the month worked	Current

Note. The last four checks are in Section 2.4 of the notebook; the others are in Sections 1 to 4. Own elaboration based on the SSMC fixed-term staff payroll, Active Transparency.

2.4 First Observations for Candidate A

To assess Candidate A, overtime hours were summed per person and month, since the legal limit applies per employee (Law 19,104, 1991). Each person-month corresponds to one person in a given month; for example, an employee present in all 24 months contributes 24 person-months. The staff category of each person-month is taken from the person's first record in that month. This

yielded 43,120 person-months, of which 182 record more than 40 daytime overtime hours (notebook, Section 4).

December 2025 was excluded from the analysis by staff category because of the recording error in that month's staff category column (Section 2.3). In the 23 months considered, 176 person-months exceed the 40-hour daytime limit, equivalent to 1.7% of person-months with paid daytime hours. The medical staff category has no cases above the limit. Proportionally, auxiliary staff concentrate the most cases (2.8% of their person-months with daytime hours), followed by technical (1.9%), administrative (1.5%) and professional staff (1.3%). Table 5 presents the results.

Table 5

Daytime and Night Overtime by Staff Category, SSMC Fixed-Term Staff, September 2024 – August 2026, Excluding December 2025

Staff category	Person-months	With daytime hrs	Above 40 daytime	Median daytime	Max daytime	With night hrs	Median night	P95 night	Max night
Administrative	5,642	2,679	40	21	80	1,559	20	104.0	205
Auxiliary	2,863	1,735	49	30	80	1,771	56	192.5	418
Physicians and related	7,936	293	0	20	40	266	57	105.0	207
Professional	16,805	3,678	48	18	160	2,528	15	110.0	279
Technical	8,026	2,108	39	20	80	1,755	32	120.0	321
Total	41,272	10,493	176			7,879			

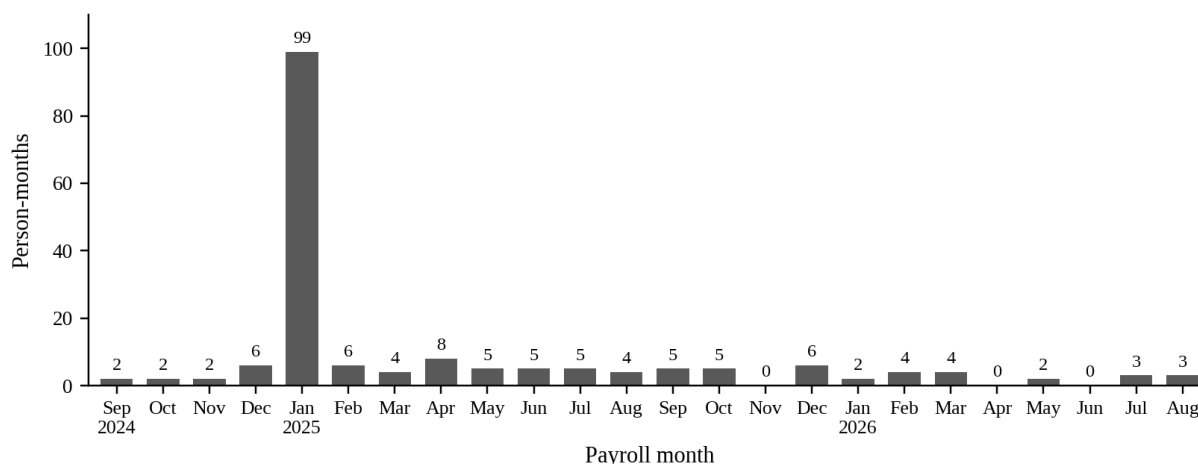
Note. Hours per person-month. Medians, 95th percentile (P95) and maxima are calculated among person-months with hours of each type. Physicians and related = physicians, pharmacists, chemist-pharmacists, biochemists and dental surgeons. Notebook, Section 4.2.

For night hours, which are not subject to that limit, auxiliary staff also show the highest values, with a median of 56 hours and a maximum of 418 hours in one month. A value of this magnitude is equivalent to almost 14 night hours per day for an entire month, which suggests that some payments may include hours from previous months. However, these data correspond to payments, not hours worked, so this cannot be verified with this source.

Cases above the limit are not evenly distributed over time either (Figure 1). Considering all 24 months, January 2025 concentrates 99 of the 182 cases (54%), while the remaining 23 months total 83 cases, with between zero and eight cases per month. January 2025 is also the only month in which the additional pay column is recorded entirely with dashes (Section 2.3). This peak could reflect a real operational need in that month or a different way of recording payments in that file, such as including hours from previous months. This source does not allow the two explanations to be distinguished.

Figure 1

Person-Months with More Than 40 Daytime Overtime Hours, by Month, SSMC Fixed-Term Staff, September 2024 – August 2026



Note. All 24 months are included, since this count does not depend on staff category. Values correspond to the notebook output in Section 4.3.

Declared entitlement to overtime was also compared with its payment. Of the 32,762 records without declared entitlement, 207 record paid hours, and of the 12,015 records with entitlement, 407 record none. The second case is expected, because being entitled does not require performing overtime. The first is an inconsistency that this source cannot explain, and it constitutes an additional criterion for prioritizing reviews (notebook, Section 2.4).

In summary, the data make it possible to measure paid daytime hours against the legal limit and to locate where atypical cases are concentrated, which supports the feasibility of the descriptive part of Candidate A. Its main limitations are that the data do not show whether the hours were worked or whether cases above the limit have an authorized exception, and that one month (January 2025) contains an atypical record that must be clarified.

2.5 First Observations for Candidate B

To assess Candidate B, overtime hours and amounts were summed per person and month, as in Candidate A. Each person was assigned their most frequent staff category over the period. With this criterion, the December 2025 recording error does not affect the classification, since that month does not change any person's predominant category (notebook, Section 5).

Persistence was measured as the percentage of months in which each person received overtime, relative to the total number of months in which the person appears in the payroll. To ensure a sufficient base of months, only the 1,776 people present for at least 12 months were considered, out of the 2,430 who appear in the period. As an operational definition, proposed by this team for this exploration and not based on a regulatory criterion, a person was considered

persistent if they received overtime in at least 75% of their months. With this criterion, 364 people (20.5%) show persistent use. Table 6 shows the results by staff category.

Table 6

Persistence of Overtime by Staff Category, People Present at Least 12 Months, SSMC Fixed-Term Staff, September 2024 – August 2026

Staff category	People	% with overtime in any month	Median % of months with overtime	Persistent people	% persistent
Administrative	239	76.2	62.5	107	44.8
Auxiliary	121	91.7	81.8	67	55.4
Physicians and related	350	6.6	0.0	9	2.6
Professional	724	46.0	0.0	118	16.3
Technical	342	59.9	8.3	63	18.4
Total	1,776			364	20.5

Note. Staff category = each person's most frequent category in the period. Persistent = overtime in at least 75% of the months present (team definition). Notebook, Section 5.2.

Auxiliary staff show the highest persistence: 91.7% received overtime at least once, the typical person received it in 81.8% of their months, and more than half (55.4%) meet the persistence criterion. Administrative staff follow, with 44.8% persistent. Among technical and professional staff, the share of persistent people is lower (18.4% and 16.3%), and the median share of months with overtime is low or zero, which indicates that in these categories use is concentrated in a subgroup of people. Nevertheless, because of the size of the category, professionals are the group with the most persistent people in absolute terms (118). The medical staff category records almost no overtime through this mechanism.

To estimate the weight of the expenditure, we calculated, for each person-month with overtime, what percentage of gross pay corresponds to the overtime amount. Three person-months with gross pay equal to zero were excluded. Table 7 presents the results.

Table 7

Overtime as a Share of Gross Pay by Staff Category, Person-Months with Overtime, SSMC Fixed-Term Staff, September 2024 – August 2026

Staff category	Person-months with overtime	Median (%)	90th percentile (%)	Maximum (%)
Administrative	2,957	12.8	28.8	112.1
Auxiliary	1,974	24.8	46.9	451.6
Physicians and related	320	20.4	32.0	37.1
Professional	4,145	11.0	30.0	120.4
Technical	2,406	15.4	32.7	199.0
Total	11,802	14.6		

Note. Overtime amount (daytime and night) as a percentage of gross pay for the month. Staff category = each person's most frequent category in the period. Notebook, Section 5.3.

In the typical month with overtime, overtime represents 14.6% of gross pay. Among auxiliary staff, the weight is considerably higher, with a median of 24.8%, and in the top 10% of cases it exceeds 46.9%.

In nine person-months, the overtime amount exceeds gross pay, with a maximum of 451.6%. This should not occur, since gross pay includes overtime. As with the extreme night-hour values observed for Candidate A, it suggests accumulated payments or recording inconsistencies that this source cannot verify.

In summary, the data make it possible to measure the persistence of overtime and its weight in pay, and they show recurring use concentrated among auxiliary and administrative staff, which supports the descriptive feasibility of Candidate B. However, persistence is compatible with a staffing gap but does not demonstrate it, because the source does not report authorized staffing or vacancies. In addition, the persistence threshold used is an exploratory definition of the team.

2.6 Proposed and Verified Joins

No joins with external sources were performed at this stage. The verified joins are internal to the payroll: name by month, to determine the unit of observation (Section 2.2); each person's staff category between consecutive months, which revealed the December 2025 error (Section 2.3); overtime amount against gross pay, which identified nine inconsistent person-months (Section 2.5); and declared overtime entitlement against paid hours, which identified 207 records with payments but no declared entitlement (Section 2.4).

Three external joins are proposed but have not been verified. First, the SSMC permanent staff payroll, published on the same portal, which would make it possible to compare the distribution of overtime between permanent and fixed-term staff, although with the same limitation of identification by name. Second, the SSMC budget glosses in the Budget Law (Law No. 21,796, 2026), which set the Service's maximum staffing and the maximum amount authorized for overtime. This join would make it possible to compare observed expenditure with the authorized ceiling (Candidate A) and observed staffing with the maximum (Candidate B), but only annually and for the Service as a whole, not by position or month. Third, the course datasets (Urgencias, REM): because the payroll does not report the facility, only an aggregate monthly join at the Service level would be possible.

3. Critical Comparison of the Candidates

3.1 Comparison Matrix

Table 8 compares both candidates along the dimensions proposed in the brief.

Table 8
Comparison of the Candidates

Dimension	Candidate A	Candidate B
Relevance supported by sources	Legal limit (Law 19,104, art. 9) and restriction in art. 95 of DFL No. 1; precedent of CGR Report No. 70/2026. No associated MINSAL priority	Victoria case; Clendon and Gibbons (2015); Scott-Marshall (2024). The National Health Strategy diagnoses human resources problems but sets no target for overtime
Fit between question and inspected data	Observed: paid daytime hours per person-month against the limit. Not observed: hours worked, Ministry of Finance exceptions and, for now, the shift allowance	Observed: months with overtime per person and its weight in pay. Not observed: authorized staffing, vacancies, leave or facility
Analytical opportunity for C1	Distribution of hours by staff category and position; analysis of January 2025; relationship between daytime and night hours; payments without declared entitlement	Persistence by staff category and position; seasonality; relationship between persistence and weight in pay; sensitivity to the threshold
Main limitation and its consequence	99 of 182 cases fall in an atypical month, so the main finding could be a recording artifact	Without staffing or vacancy data, it can only describe persistent use, not demonstrate a gap
Feasible scope and future ML/dashboard potential	Detection of atypical profiles without labels; dashboard for internal audit (Section 4.1)	Prediction of persistence from payment history; dashboard for people management (Section 4.2)

Note. Own elaboration based on Sections 1 and 2.

3.2 Provisional Preference

The team provisionally prefers Candidate B, for three reasons. First, its signal is broader and more stable: 364 people show persistent use, whereas in Candidate A, excluding January 2025, the cases above the limit total 83 over 23 months, with a maximum of eight per month. Second, it is less sensitive to the quality problems detected: the December 2025 staff category error does not alter the classification by predominant category, whereas the main result of Candidate A depends on a month whose record cannot be explained with this source. Third, it uses the full 24-month series, which offers a better basis for a machine learning extension (Section 4).

The main uncertainty is that the source does not report authorized staffing or vacancies, so Candidate B cannot demonstrate a staffing gap. If it is developed in C1, the question should be framed in terms of persistent use rather than a gap. The 40-hour daytime limit from Candidate A can be incorporated as a complementary indicator.

Two findings could change this preference: that the special allowance codes identify the shift allowance, which would allow Candidate A to apply the restriction in Article 95, or that January 2025 corresponds to a real operational need rather than an atypical record. Feedback indicating that Candidate B is not sufficient for C1 without staffing data would also change it.

3.3 Questions for Feedback

1. Is it sufficient for Candidate B to conclude that persistent use is compatible with a staffing gap, without being able to demonstrate it, or should a staffing source be incorporated in C1?

2. Given that the payroll does not include the RUT, is it appropriate to identify people by full name, or would it be better to work only at the level of records and positions?
3. For January 2025 and December 2025, do you recommend excluding them, analyzing them separately, or requesting clarification from the SSMC through a freedom of information request?
4. Is a join with the course datasets (Urgencias, REM) expected, considering that it would only be possible at the aggregate monthly level of the Service?

4. Future Machine Learning and Dashboard Potential

4.1 Candidate A

A future machine learning question would be to identify, from the payroll variables (staff category, grade, daytime and night hours, and the weight of overtime in pay), which payments show an atypical profile relative to others in the same staff category. Because the source does not indicate which payments were objected to, there are no labels for a supervised approach, and alerts could only be validated with attendance records or audit results, which are not public. It remains to be verified whether the special allowance codes identify the shift allowance. As this team's proposal, a dashboard for the internal audit unit could show, with each published payroll, the person-months above 40 daytime hours by staff category, their monthly evolution and the months with atypical records, in order to decide which authorizations to review first. Its usefulness is limited by the small number of cases outside January 2025.

4.2 Candidate B

A future question would be to predict whether a staff category or position will continue to receive overtime payments in the following months, based on its history in the payroll. The 24-month series makes it possible to build that history, but it covers only two annual cycles, and the lack of a unique identifier makes tracking people difficult. Two elements remain to be verified: whether the observations column reliably identifies replacements and substitutions, and whether there is a source of staffing or vacancies by position that would make it possible to relate persistence to a gap. As this team's proposal, a dashboard for the people management and development units could show the presence of overtime by staff category and month, the share of people with persistent use and the weight of overtime in gross pay, in order to decide in which staff categories to assess a staffing adjustment first.

5. Conclusions

This report explored two candidate lines of inquiry on the use of overtime among SSMC fixed-term staff, based on 24 monthly Active Transparency payrolls (44,777 records). The inspection showed that the source makes it possible to address the descriptive part of both questions, subject to three conditions: each row corresponds to a contract record rather than a person, identification depends on the full name, and two months contain atypical records. In

Candidate A, 182 person-months exceed the 40-hour daytime limit, but 99 of them are concentrated in January 2025. In Candidate B, 20.5% of people present for at least 12 months show persistent use, concentrated among auxiliary and administrative staff, and in the typical month with overtime it represents 14.6% of gross pay. Neither candidate makes it possible to verify whether the hours paid were worked. The team provisionally prefers Candidate B, subject to feedback on the questions in Section 3.3.

6. Process Record

6.1 Decisions and Checks

Table 9

Decisions and Checks

Date / stage	Decision	Alternative considered	Reason and verification	Location
24-09-2026 / Part A	Frame the problem as atypical payments, not fraud	Original candidate on fraudulent overtime	The data record payments, not hours worked. Verified in CGR Report No. 70/2026, which had to use attendance records	Section 1.1
24-09-2026 / Part A	Apply the 40-hour limit only to daytime hours	Apply it to total hours	Art. 9 of Law 19,104 and FALMED	Section 1.2.2
24-09-2026 / Part A	Cite the original CGR report instead of the press	Cite news articles	The brief requires following news to the official source. Figures cross-checked: they match	Section 1.2.2; References
24-09-2026 / Part A	Identify the organization (SSMC) and its network	—	Official SSMC website	Section 1.1
25-09-2026 / Part B	Read all columns as text, latin-1, ";" separator	Automatic type inference	Columns with mixed formats ("44 Hrs.", "\$")	Section 2.1; notebook §1
25-09-2026 / Part B	Treat "-" as missing	Treat as zero	The source does not state its meaning	Sections 2.1, 2.3; notebook §2.1
25-09-2026 / Part B	Code "No tiene" as zero	Treat as missing	It indicates no paid hours	Section 2.1; notebook §2.2
25-09-2026 / Part B	Sum hours per person-month	Analyze by row	The limit applies per employee; 57–68 people per month have more than one record	Sections 2.2, 2.4; notebook §3–4
25-09-2026 / Part B	Staff category of a person-month = first record of the month	Other record of the month	Needed to assign one category per person-month	Section 2.4; notebook §4.1
25-09-2026 / Part B	Exclude December 2025 from analyses by category	Keep or correct it	218 and 213 changes versus six or fewer in other months	Section 2.3; notebook §4.1
25-09-2026 / Part B	Use predominant staff category in Candidate B	Month-by-month category	Neutralizes the December 2025 error	Section 2.5; notebook §5.1

Date / stage	Decision	Alternative considered	Reason and verification	Location
25-09-2026 / Part B	Persistence: $\geq 75\%$ of months and ≥ 12 months present	Other thresholds	Exploratory definition, not regulatory	Section 2.5; notebook §5.2
25-09-2026 / Part B	Exclude 3 person-months with zero gross pay	Include them	Division by zero	Section 2.5; notebook §5.3
25-09-2026 / Part B	Do not use position as a proxy for the facility	Use position as a proxy	95 generic values; 81 after unifying case	Section 2.2; notebook §2.4
25-09-2026 / Part B	Do not state which code is the shift allowance	Assume a code	No legend for the codes in the file	Section 1.2.4; notebook §2.4
25-09-2026 / Part C	Provisionally prefer Candidate B	Candidate A	Broader, more stable signal; less sensitive to quality problems	Section 3.2

Note. Dates in the format dd-mm-yyyy.

6.2 Instructor Feedback and Response

Table 10

Instructor Feedback

Advice received and date	Response	Reason	Change and location
On 24-09-2026, the instructor indicated that the team could work on its own topic or any other, following the same structure	Accepted	It allows the team to keep the overtime topic	Candidate A is presented as a public management problem, without forcing a link to MINSAL (Section 1.2.2)
Feedback on the questions in Section 3.3	Pending	—	Provisional decision: Candidate B (Section 3.2)

6.3 Individual Contributions

Table 11

Individual Contributions

Member	Concrete contribution	Location of evidence
Alessandro Lavezzi	Data analysis. Report.	Github.
José Saavedra	Data aquisition. Data Analysis. Report.	Github.

Note. Actual contributions are recorded, not only the planned distribution of tasks.

6.4 Use of Artificial Intelligence and Verification

Table 12

Use of Artificial Intelligence

Tool and purpose	Relevant excerpt	Accepted, modified or rejected	Verification
Claude (Anthropic): interpret the brief and propose candidates	Prompt: "Give us the two candidates according to our topic." Output: "The data cannot prove fraud. They show how many hours were paid, not whether they were worked, so the problem should be reframed as atypical or above-limit overtime, not as fraud."	Modified: the candidate was reframed from "fraud" to atypical payments	Checked against the brief and CGR Report No. 70/2026
Claude: search for and validation of sources	Prompt: "Validate by searching the internet." Output: "The 40-hour limit applies only to daytime overtime. The 229, 221 and 218 night hours we saw do not violate that limit."	Modified: corrected so that the 40-hour limit applies only to daytime hours	Reading of Law 19,104 (BCN) and FALMED

6.5 Individual Reflection

Alessandro Lavezzi.

I reviewed the draft report against the brief and the notebook outputs, and ran the additional checks in Section 2.4 of the notebook (position values, overtime entitlement versus payment, and observation and allowance codes). The main learning was that a field's name does not guarantee what it measures: the position column looked like a possible proxy for the facility, but its 95 values turned out to be generic functions, and the entitlement field contradicted actual payments in 207 records. An unresolved uncertainty is whether any of the special allowance codes identifies the shift allowance; if it does, Candidate A could test the Article 95 restriction, which might change our preference.

José Saavedra.

I drafted Section 1 of the report, including the documentary foundation for both candidates, and built the notebook's loading, conversion and analysis steps (Sections 1 to 5). The main learning was that a payroll row is not a person: because some employees hold more than one contract in the same month, hours had to be summed per person-month before comparing them with the legal limit, otherwise the count of cases above 40 hours would have been wrong. I also learned to follow news coverage back to the original Comptroller report before citing any figure. An unresolved uncertainty is whether the January 2025 peak, which concentrates 99 of the 182 cases above the limit, reflects a real operational need or a different way of recording payments in that file; this source cannot distinguish between the two.

References

- BioBioChile. (2026, July 27). *Servicio de Salud descarta mal uso de recursos en Hospital de Victoria tras informe de Contraloría*.
<https://www.biobiochile.cl/noticias/nacional/region-de-la-araucania/2026/07/27/servicio-de-salud-descarta-mal-uso-de-recursos-en-hospital-de-victoria-tras-informe-de-contraloria.shtml>
- Clendon, J., & Gibbons, V. (2015). 12 h shifts and rates of error among nurses: A systematic review. *International Journal of Nursing Studies*.
<https://www.sciencedirect.com/science/article/abs/pii/S0020748915000735>
- Consejo para la Transparencia. (2022). *Fiscalización focalizada: horas extras del sector municipal*. https://www.consejotransparencia.cl/wp-content/uploads/2022/06/Informe-HH.EE-2022_07-06-2022-FINAL-rev-hm-1.pdf
- Contraloría General de la República, Contraloría Regional de La Araucanía. (2026, July 14). *Informe Final de Investigación Especial N°70, de 2026, sobre trabajos extraordinarios en el Hospital San José de Victoria*.
<https://www.contraloria.cl/buscadorpdf/auditoria/78e998a9d6e1444630fbd1d64116ee06/html>
- Decreto con Fuerza de Ley N°1. (2005). Ministerio de Salud. Articles 94, 95 and 97, cited in Contraloría General de la República (2026).
- Diario Constitucional. (2025, November 30). *Contraloría aclara criterios para pago de horas extraordinarias durante períodos de ausencia*.
<https://www.diarioconstitucional.cl/2025/11/30/contraloria-aclara-criterios-para-pago-de-horas-extraordinarias-durante-periodos-de-ausencia/>
- FALMED, Unidad de Defensa Laboral Médica. (n.d.). *¿Hay algún límite para el establecimiento del máximo de horas extraordinarias?* Retrieved September 24, 2026, from <https://falmed.cl/udelam/materia-jornada-trabajo/hay-algun-limite-para-el-establecimiento-del-maximo-de-horas>
- Inteligencia Jurídica. (2025, March). *CIC: Contraloría analiza los pagos de horas extras en los municipios a nivel nacional*. <https://www.portal.ijuridica.cl/2025/03/cic-contraloria-analiza-los-pagos-de-horas-extras-en-los-municipios-a-nivel-nacional/>
- Las Noticias de Malleco. (2026, July 28). *Directora de Hospital San José de Victoria se refiere a informe de Contraloría*. <https://lasnoticiasdemalleco.cl/salud/directora-de-hospital-san-jose-de-victoria-se-refiere-a-informe-de-contraloria/>
- Ley N°18.834. Estatuto Administrativo. Articles 65, 66 and 69, cited in Contraloría General de la República (2026).

- Ley N°19.104. (1991). *Reajusta remuneraciones de los trabajadores del sector público*. Biblioteca del Congreso Nacional. <https://www.bcn.cl/leychile/navegar?idNorma=30471>
- Ley N°21.796. (2026). *Ley de Presupuestos del Sector Público correspondiente al año 2026*. Biblioteca del Congreso Nacional. <https://www.bcn.cl/leychile/navegar?idNorma=1219410>
- Ministerio de Salud. (2022). *Estrategia Nacional de Salud para los Objetivos Sanitarios al 2030*. <https://www.minsal.cl/wp-content/uploads/2022/03/Estrategia-Nacional-de-Salud-2022-MINSAL-V8.pdf>
- Ministerio de Salud. (2026). *Estrategia Nacional de Salud 2021-2030* (updated version). <https://www.minsal.cl/wp-content/uploads/2026/04/Estrategia-Nacional-de-Salud-2021-2030.pdf>
- Scott-Marshall, H. K. (2024). Safe limits on work hours for the nursing profession: A rapid evidence review. *Frontiers in Global Women's Health*, 5. <https://doi.org/10.3389/fgwh.2024.1455422>
- Servicio de Salud Metropolitano Central. (n.d.-a). *Quiénes somos*. Retrieved September 24, 2026, from <https://ssmc.gob.cl/>
- Servicio de Salud Metropolitano Central. (n.d.-b). *Red de atención*. Retrieved September 24, 2026, from <https://ssmc.redsalud.gob.cl/centros-de-salud/>
- Servicio de Salud Metropolitano Central. (n.d.-c). *SAMU*. Retrieved September 24, 2026, from <https://ssmc.gob.cl/centros-de-salud/samu/>
- Servicio de Salud Metropolitano Central. (2024–2026). *Personal a contrata*. Portal de Transparencia del Estado. Downloaded September 15, 2026, from <https://www.portaltransparencia.cl/PortalPdT/>
- Servicio de Salud Metropolitano Central. (2025, December 20). *SAMU Metropolitano conmemora 30 años de existencia*. <https://ssmc.gob.cl/samu-metropolitano-conmemora-30-anos-de-existencia-con-inauguracion-de-nuevo-espacio-y-reconocimiento-a-funcionarios/>
- vLex. (n.d.). *Horas extraordinarias diurnas*. Retrieved September 24, 2026, from <https://vlex.cl/tags/horas-extraordinarias-diurnas-2225872>

Appendices

Appendix A. Field Dictionary of the Fixed-Term Staff Payroll

Table A1
Columns of the Monthly Payroll File

#	Column (original name)	Format in the file	Description and observations
1	Año	Numeric text	Year of pay
2	Mes	Text	Month of pay, in words
3	Estamento	Categorical	Staff category; 5 categories; recording error in December 2025
4	Nombre completo	Text	"SURNAMES, NAMES"; only available identifier
5	Cargo o función	Free text	Position or function; 95 values (81 after unifying case); generic functions
6	Grado EUS o jornada	Mixed text	Pay grade ("15") or working hours ("44 Hrs.")
7	Calificación profesional o formación	Free text	Degree or training
8	Región	Text	Region of employment
9	Asignaciones especiales del mes	Codes in parentheses	Special allowances; included in gross pay; no legend
10	Remuneración bruta del mes	Amount (\$)	Gross pay; includes bonuses, allowances and overtime
11	Remuneración líquida del mes	Amount (\$)	Net pay
12	Rem. adicionales del mes	Amount (\$)	Additional pay; not included in gross pay; "-" in January 2025
13	Remuneración Bonos incentivos del mes	Amount (\$)	Bonuses; included in gross pay; "-" in four months
14	Derecho a horas extraordinarias	Sí / No	Entitlement to overtime
15–17	Montos y horas extraordinarias diurnas / nocturnas / festivas	"\$ amount : hours hrs" or "No tiene"	Daytime, night and holiday overtime; holiday: "No tiene" in every record
18–19	Fecha de inicio / término	dd/mm/yyyy	Contract start and end dates; header says dd/mm/aa
20	Viáticos del mes	Amount (\$)	Travel allowances; not included in gross pay; not inspected
21	Observaciones	Free text	Observations; 1,921 mentions of replacement or substitution

Note. All columns are read as text. A final unnamed empty column (generated by the trailing semicolon) is dropped. Notebook, Sections 1 and 2.

Appendix B. Supplementary Inspection Tables

Table B1

Rows, Distinct People and Repeated Records by Month

Month	Rows	Distinct people	Additional rows	People with more than one record
2024-09	1,781	1,717	64	60
2024-10	1,799	1,733	66	62
2024-11	1,798	1,734	64	61
2024-12	1,803	1,739	64	60
2025-01	1,825	1,756	69	63
2025-02	1,842	1,776	66	60
2025-03	1,845	1,783	62	57
2025-04	1,861	1,798	63	59
2025-05	1,874	1,804	70	66
2025-06	1,870	1,795	75	67
2025-07	1,872	1,799	73	65
2025-08	1,884	1,811	73	66
2025-09	1,878	1,811	67	63
2025-10	1,882	1,806	76	66
2025-11	1,919	1,844	75	68
2025-12	1,918	1,848	70	65
2026-01	1,907	1,838	69	62
2026-02	1,951	1,880	71	64
2026-03	1,908	1,839	69	62
2026-04	1,903	1,829	74	65
2026-05	1,886	1,814	72	63
2026-06	1,865	1,795	70	63
2026-07	1,857	1,788	69	61
2026-08	1,849	1,783	66	59

Note. People are identified by full name. Notebook, Section 3.

Table B2

Staff Category Changes Relative to the Previous Month

Month	People changing category
2024-12	1
2025-01	2
2025-02	1
2025-09	1
2025-11	1
2025-12	218

Month	People changing category
2026-01	213
2026-02	6
2026-08	1

Note. Months not listed have no changes. Notebook, Section 4.1.

Table B3

Person-Months with More Than 40 Daytime Overtime Hours, by Month

Month	Cases	Month	Cases	Month	Cases
2024-09	2	2025-05	5	2026-01	2
2024-10	2	2025-06	5	2026-02	4
2024-11	2	2025-07	5	2026-03	4
2024-12	6	2025-08	4	2026-04	0
2025-01	99	2025-09	5	2026-05	2
2025-02	6	2025-10	5	2026-06	0
2025-03	4	2025-11	0	2026-07	3
2025-04	8	2025-12	6	2026-08	3
Total					182

Note. Data for Figure 1. Notebook, Section 4.3.